MODULE 1 · LECTURE 15

Geospatial Technology in Agriculture: Concepts, Techniques and Applications

Remote sensing, satellite positioning and geographic information systems — how crops are seen from space, how vegetation indices and classified images are made and checked, how location is measured, and how maps support forecasting, insurance and precision farming.

Learning objectives

After this lecture you should be able to:

1. Define geospatial technology and describe its three components: remote sensing, GNSS and GIS.
2. Explain the principles of remote sensing, spectral signatures and the four types of resolution.
3. Calculate NDVI and interpret it for crop condition.
4. Describe image classification and assess its accuracy with a confusion matrix.
5. Explain how GNSS determines position and describe NavIC.
6. Describe GIS data models, coordinate systems, map scale and common GIS operations.
7. Discuss agricultural applications of geospatial technology in India.

Scope: Lectures 11 and 13 used satellite data for forecasts and water accounting, and Lecture 12 used GNSS for variable-rate machines. This lecture explains the technology itself. Decision support systems follow in Lecture 16, big data in Lecture 23, and yield mapping and AI-based precision agriculture in Lecture 26.

1. What is geospatial technology?

Geospatial technology collects, stores, analyses and displays information that is tied to a location on the Earth. In agriculture it answers questions such as: where is wheat grown this season, which fields are under stress, how much area is affected by drought, and how should fertiliser vary within a field? It has three main components.

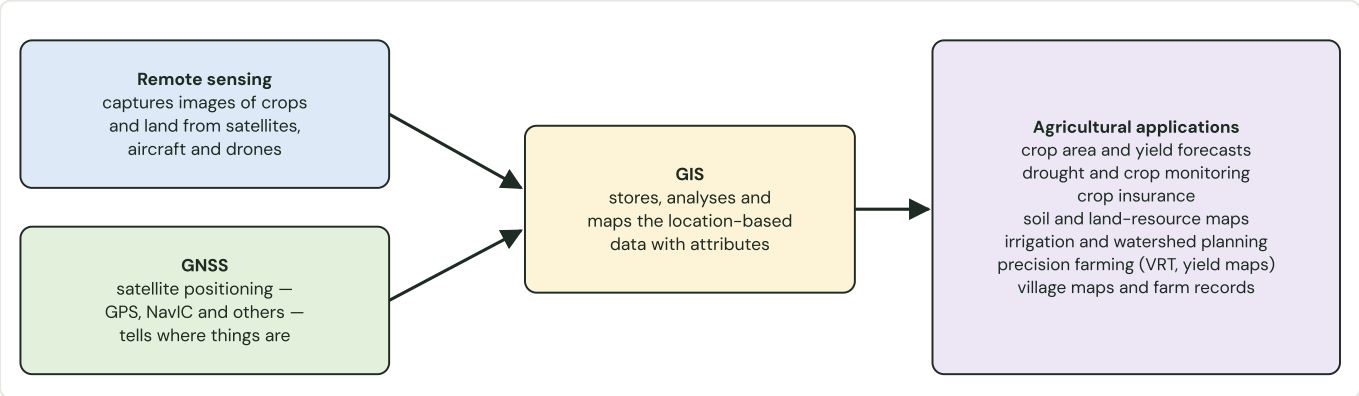


Figure 1. The three components of geospatial technology and their uses in agriculture. Original diagram.

2. Remote sensing

Remote sensing is acquiring information about objects without touching them, by measuring the electromagnetic radiation they reflect or emit. **Passive** sensors record sunlight reflected by the surface (optical sensors) or heat emitted by it (thermal sensors). **Active** sensors send out their own energy and

record what returns — **radar** (synthetic aperture radar, SAR) works day and night and through clouds, which is valuable in the monsoon season.

2.1 Spectral signatures

Each surface reflects different wavelengths differently — its **spectral signature**. Healthy green leaves absorb most blue and red light for photosynthesis, reflect some green (which is why they look green) and reflect strongly in the **near-infrared (NIR)** because of their internal leaf structure. A stressed crop reflects more red and less NIR; dry soil reflects moderately and rises gradually with wavelength; water reflects very little, especially in the NIR.

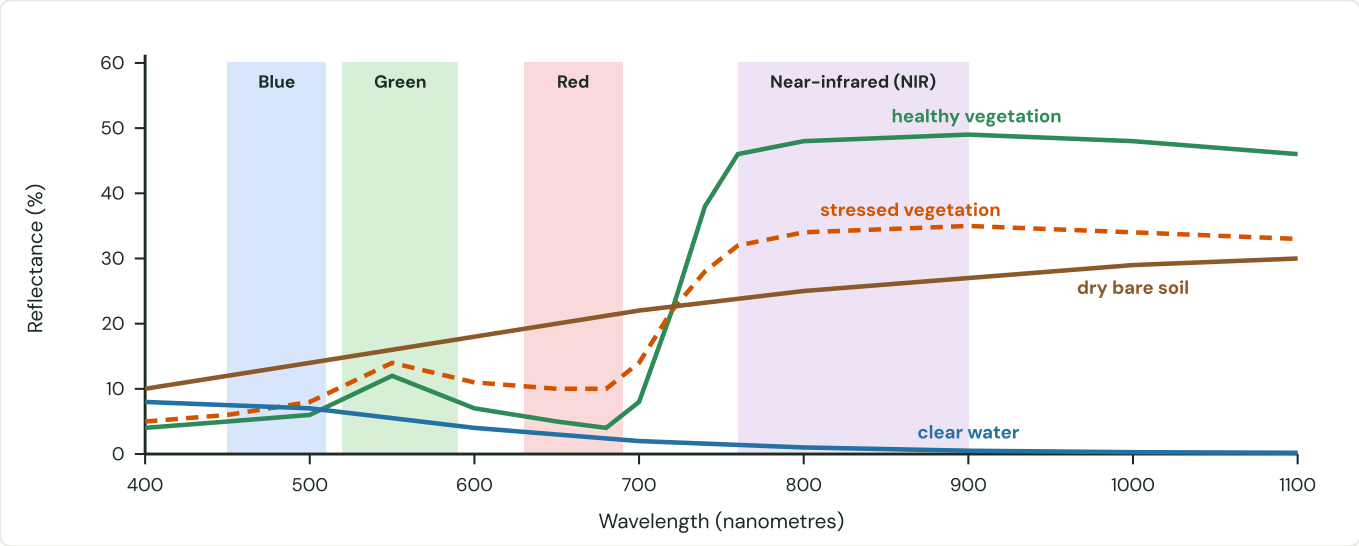


Figure 2. Typical spectral reflectance curves (schematic; values vary with crop, soil and conditions). Original chart.

2.2 Resolution and platforms

Resolution	Meaning	Example
Spatial	Size of the ground area represented by one pixel	10 m pixels show individual fields; 56 m pixels mix several small fields
Spectral	Number and width of wavelength bands recorded	Sentinel-2 records 13 bands
Temporal	How often the same place is imaged (revisit time)	Every 5 days for the two Sentinel-2 satellites at the Equator
Radiometric	Number of brightness levels the sensor distinguishes	10-bit data have 1,024 levels; 12-bit data 4,096

Satellite / sensor	Spatial resolution	Notes
Sentinel-2 (European Union / ESA, MSI)	10 m (4 bands), 20 m (6 bands), 60 m (3 bands)	13 bands, 290 km swath, 5-day revisit with two satellites; data free (European Space Agency [ESA], n.d.)
Resourcesat-2 and 2A (ISRO) — LISS-IV	5.8 m	High-resolution multispectral camera with 5-day revisit capability (Indian Space Science Data Centre [ISSDC], n.d.)
Resourcesat — LISS-III	23.5 m	Medium resolution; with two satellites, revisit improved to 12–13 days (ISSDC, n.d.)
Resourcesat — AWiFS	56 m	Wide field; with two satellites, revisit improved to 2–3 days; used for large-area crop monitoring (ISSDC, n.d.)
Drones (UAVs)	A few centimetres	Field-level detail on demand; small areas

Worked example 1 — choosing a resolution for small fields
Pixel area = (pixel size)². Sentinel-2: 10 × 10 = 100 m² → 100 pixels per hectare. LISS-IV: 5.8² = 33.64 m² → 297 pixels/ha. LISS-III: 23.5² = 552 m² → 18 pixels/ha. AWiFS: 56² = 3,136 m² → 3.2 pixels/ha.
A typical 0.4-ha field therefore covers 40 Sentinel-2 pixels, 119 LISS-IV pixels, 7 LISS-III pixels — but only about 1.3 AWiFS pixels. Most AWiFS pixels over small farms are **mixed pixels** containing several fields, so field-level mapping needs 10 m or finer data, while AWiFS suits district- and state-level monitoring.

2.3 Vegetation indices

Because healthy crops are dark in red and bright in NIR, the difference between the two bands is a sensitive indicator of green biomass and vigour. The most widely used index, the **Normalised Difference Vegetation Index (NDVI)**, follows from red and NIR combinations shown by Tucker (1979) to track green vegetation:

$$\mathrm{NDVI} = (\mathrm{NIR} - \mathrm{Red}) \div (\mathrm{NIR} + \mathrm{Red}) \qquad \text{range } -1 \text{ to } +1$$

Worked example 2 — NDVI from reflectance (illustrative values)
Healthy crop: NIR 0.50, red 0.08 → (0.50 - 0.08) ÷ (0.50 + 0.08) = 0.72
Stressed crop: NIR 0.35, red 0.12 → 0.23 ÷ 0.47 = 0.49
Bare soil: NIR 0.30, red 0.25 → 0.05 ÷ 0.55 = 0.09
Water: NIR 0.02, red 0.05 → -0.03 ÷ 0.07 = -0.43
Dense healthy vegetation gives high positive values; soil gives values near zero; water gives negative values. A fall in NDVI over a field between two dates signals stress worth checking on the ground.

2.4 From image to map

1. **Pre-processing** — correct for sensor and atmospheric effects (radiometric correction) and fit the image to map coordinates (geometric correction).
2. **Enhancement** — combine bands into colour composites and compute indices such as NDVI.
3. **Classification** — assign each pixel to a class (wheat, mustard, fallow, water…). In **supervised** classification the analyst supplies training samples of known classes; in **unsupervised** classification the computer groups similar pixels and the analyst labels the groups. Machine-learning classifiers are now common (Lectures 19–28).
4. **Accuracy assessment** — compare the map with ground-truth points in a **confusion (error) matrix** (Congalton, 1991).

5. **Area estimation and reporting** — count classified pixels × pixel area, and publish maps and statistics.

Worked example 3 — accuracy of a crop map and crop area (illustrative data). A crop map was checked at 250 ground-truth points:

Map class ↓ / Ground truth →	Wheat	Mustard	Other	Row total
Wheat	**88**	6	4	98
Mustard	7	**52**	3	62
Other	5	2	**83**	90
Column total	100	60	90	250

Overall accuracy = (88 + 52 + 83) ÷ 250 = 223 ÷ 250 = 89.2%.
Wheat — producer's accuracy (ground wheat correctly mapped) = 88 ÷ 100 = 88.0%; user's accuracy (mapped wheat that really is wheat) = 88 ÷ 98 = 89.8%.
Kappa = (0.892 − 0.346) ÷ (1 − 0.346) = 0.835, where 0.346 = (98×100 + 62×60 + 90×90) ÷ 250² is the agreement expected by chance.
If 1,850,000 Sentinel-2 pixels in a district are classified as wheat: area = 1,850,000 × 100 m² = 18,500 ha (before correcting for the map's errors).

3. Global navigation satellite systems (GNSS)

A GNSS receiver calculates its position from signals broadcast by satellites. Each signal carries the satellite's position and the time it was sent; the travel time gives the distance. With distances to at least four satellites, the receiver solves for latitude, longitude, height and its own clock error.

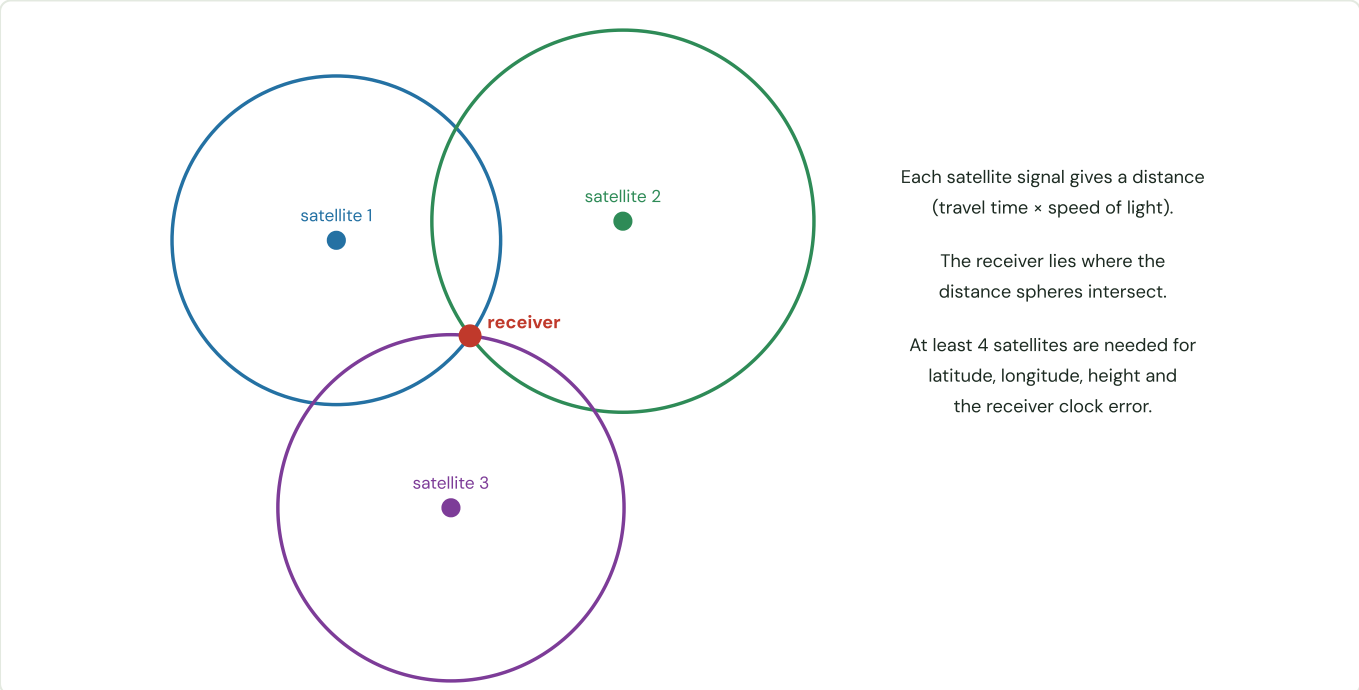


Figure 3. The principle of satellite positioning (shown in two dimensions). Original diagram.

System	Operated by	Coverage
GPS	United States	Global
GLONASS	Russia	Global
Galileo	European Union	Global
BeiDou	China	Global
NavIC (formerly IRNSS)	India (ISRO)	India and up to 1,500 km beyond its boundary

NavIC is designed with a constellation of seven satellites — three in geostationary orbit and four in inclined geosynchronous orbit — and provides a Standard Positioning Service for civilian users with position accuracy better than 20 m (Indian Space Research Organisation [ISRO], n.d.-b). Ordinary receivers give accuracy of a few metres; **differential GNSS** and **real-time kinematic (RTK)** correction from a nearby base station improve this to centimetres, as needed for auto-steer and controlled-traffic farming (Lecture 12).

- Mapping field boundaries and areas; geo-tagging soil-sample points and crop photos (Lecture 14).
- Guidance and auto-steer of tractors; section control and variable-rate application (Lecture 12).
- Yield monitoring and mapping on combine harvesters (Lecture 26).
- Locating farms, warehouses, mandis and service providers in apps.

4. Geographic information systems (GIS)

A **GIS** is a computer system for capturing, storing, analysing and displaying spatial data together with their **attributes** — the descriptive data in a linked database table (Lecture 6). It has five components: hardware, software, data, people and methods.

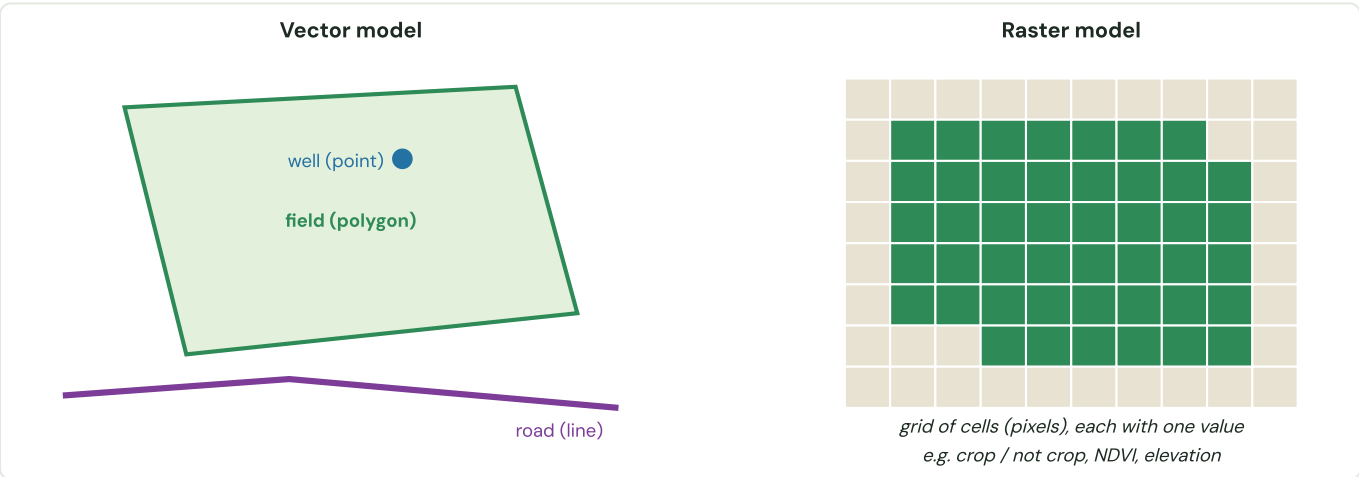


Figure 4. The same landscape in the vector and raster data models. Original diagram.

	Vector model	Raster model
Represents features as	Points, lines and polygons	A grid of cells (pixels), each with a value
Best for	Discrete features with clear boundaries — fields, roads, wells, village boundaries	Continuous surfaces — satellite images, NDVI, elevation, rainfall, soil moisture
Advantages	Precise boundaries; compact; attribute tables	Simple structure; easy overlay and map algebra
Limitations	Complex for continuous data	Large files; boundaries limited by cell size

Coordinates and map scale

- **Geographic coordinates** give latitude and longitude in degrees; the WGS 84 datum is used by GPS. **Projected coordinates** such as UTM give positions in metres on a flat map, convenient for measuring distance and area.
- One degree of latitude is about 111 km, so 0.001° of latitude is about 111 m.
- **Map scale** relates map distance to ground distance: at 1 : 50,000, 1 cm on the map is 50,000 cm = 500 m on the ground.

> **Worked example 4 — coordinates and scale**
> Convert 26° 54′ 36″ N to decimal degrees: 26 + 54/60 + 36/3600 = 26.91° N.
> On a 1 : 50,000 map, a block measuring 2 cm × 3 cm covers 1,000 m × 1,500 m on the ground = 1.5 km² = 150 ha.

GIS operations

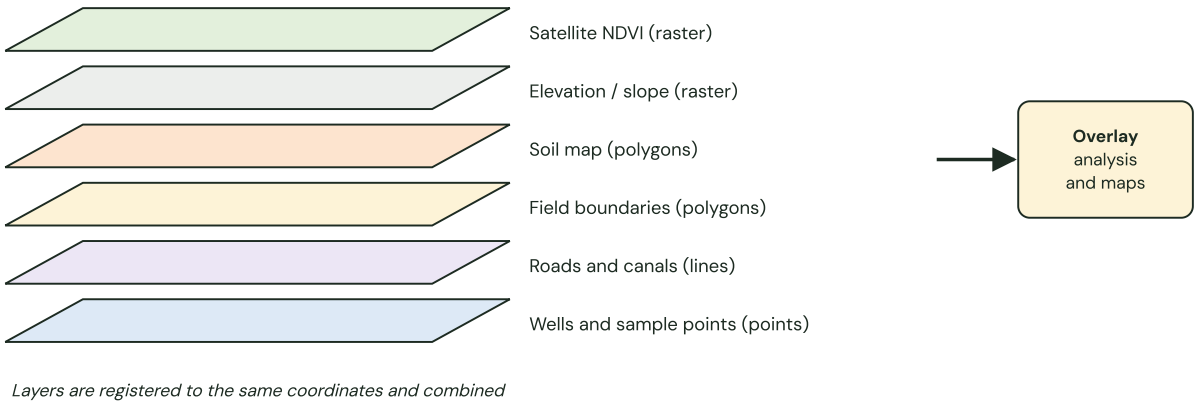


Figure 5. Thematic layers combined in a GIS. Original diagram.

Operation	What it does	Agricultural example
Overlay	Combines layers to find where conditions coincide	Fields on sloping, low-fertility soil
Buffer	Creates a zone of set width around features	50-m spray-free zone around wells and canals
Spatial query	Selects features by location or attribute	All plots within 5 km of a mandi with soil N below a threshold
Interpolation	Estimates values between sample points (e.g. inverse-distance weighting, kriging)	Continuous soil-nutrient map from grid soil samples
Map algebra	Cell-by-cell calculation on rasters	Computing NDVI for every pixel
Terrain analysis	Slope, aspect and drainage from elevation data	Planning bunds, drains and watershed works
Network analysis	Shortest or cheapest routes along networks	Routing produce from villages to warehouses

Software: QGIS is free and open-source GIS software licensed under the GNU GPL (QGIS Project, n.d.); commercial packages are also widely used. **Bhuvan**, the Indian geo-platform of ISRO run by the National Remote Sensing Centre, provides Indian satellite images, thematic maps and services online (National Remote Sensing Centre [NRSC], n.d.).

5. Applications in agriculture

Application	How geospatial technology is used	Indian example
Crop acreage and production forecasting	Classified satellite images give crop area; weather and spectral indices give yield	Pre-harvest forecasts of nine field crops by the Mahalanobis National Crop Forecast Centre under FASAL (ISRO, n.d.-a; Lecture 11)
Drought assessment	Vegetation indices and weather data track crop condition through the season	Operational agricultural drought assessment by MNCFC (ISRO, n.d.-a)
Crop insurance	Satellite data, weather data and models estimate yields, reducing dependence on manual crop-cutting experiments	YES-TECH, introduced in 2023 and implemented by twelve states with a minimum 30% weightage in yield estimation (Mahaur, n.d.)
Digital farm records	Geo-referenced maps link farmers, plots and crops	Geo-referenced Village Maps and the Crop Sown Registry under AgriStack (Press Information Bureau [PIB], 2024; Lecture 31)
Soil and land-resource mapping	Satellite data, terrain and soil samples combined in GIS	Soil, land-capability and land-use maps
Irrigation and watershed planning	Elevation, slope, drainage and land-use layers	Locating check dams and farm ponds; command-area monitoring
Precision agriculture	Management zones and prescription maps from soil, yield and canopy data; GNSS-guided application	Variable-rate fertiliser and seed (Lectures 12 and 26)

Remote sensing has advanced precision agriculture by mapping within-field variation in crops and soils, but gaps remain in turning images into reliable management decisions (Mulla, 2013).

Under YES-TECH, satellite-based estimates are combined with **crop-cutting experiments (CCEs)** — small plots chosen at random in a notified area are harvested and weighed, and the actual yield is compared with the threshold yield to decide insurance claims. CCEs are resource-intensive and open to error, which is why remote sensing is being added (Mahaur, n.d.).

Benefits	Limitations
Repeated, objective coverage of large areas	Clouds block optical images during the monsoon (radar helps)
Early warning of stress, drought and pests	Small, mixed fields need high-resolution data
Less manual survey work; transparent records	Needs ground truth for training and validation
Free data from several satellites; free software	Skills and computing capacity required
Supports precise, field-level decisions	Privacy and accuracy of farm-level maps must be safeguarded

6. Key takeaways

- Geospatial technology combines remote sensing (seeing), GNSS (locating) and GIS (analysing and mapping).
- Healthy vegetation absorbs red and reflects strongly in near-infrared; NDVI = (NIR - Red) ÷ (NIR + Red) captures this.
- Spatial, spectral, temporal and radiometric resolution decide what a sensor can detect; small Indian fields need 10 m or finer pixels.
- Sentinel-2 offers free 10-m, 5-day data; Resourcesat carries LISS-IV (5.8 m), LISS-III (23.5 m) and AWiFS (56 m).
- Classified maps must be checked with a confusion matrix — overall, producer's and user's accuracy and kappa.
- GNSS positioning needs at least four satellites; NavIC (seven satellites) serves India and 1,500 km beyond with better than 20 m accuracy.
- GIS stores vector and raster data with attributes and performs overlay, buffer, interpolation and map algebra; QGIS and Bhuvan are freely accessible.
- Applications include crop forecasting (FASAL), drought assessment, crop insurance (YES-TECH), village maps (AgriStack) and precision farming.

7. Examination questions

Short-answer questions (2 marks)

Q1. Define remote sensing.

Acquiring information about objects without contact, by measuring the electromagnetic radiation they reflect or emit.

Q2. Distinguish passive and active sensors.

Passive sensors record reflected sunlight or emitted heat; active sensors (radar) send their own energy and record the return.

Q3. Why does healthy vegetation reflect strongly in the near-infrared?

Because of the internal structure of healthy leaves; chlorophyll absorbs red and blue light.

Q4. Write the NDVI formula and its range.

NDVI = (NIR - Red)/(NIR + Red); from -1 to +1.

Q5. Name the four types of resolution.

Spatial, spectral, temporal and radiometric.

Q6. State the spatial resolutions of LISS-IV, LISS-III and AWiFS.

5.8 m, 23.5 m and 56 m.

Q7. Why does a GNSS receiver need at least four satellites?

To solve for latitude, longitude, height and the receiver's clock error.

Q8. What is NavIC?

India's regional navigation satellite system of seven satellites, covering India and 1,500 km beyond, with better than 20 m accuracy.

Q9. Distinguish vector and raster data models.

Vector uses points, lines and polygons; raster uses a grid of cells with values.

Q10. What is a buffer in GIS?

A zone of specified width around a feature, e.g. a spray-free zone around a well.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the principles of remote sensing and the spectral signatures of vegetation, soil and water.

EMR, passive and active sensors, Figure 2, reasons for each curve; radar and clouds.

Q2. Describe the types of resolution and the satellites used for agricultural remote sensing.

Resolution table; Sentinel-2 and Resourcesat sensors; drones; worked example 1 on pixel size and small fields.

Q3. Explain vegetation indices and calculate NDVI for given reflectances.

Red/NIR behaviour; NDVI formula; worked example 2; interpretation.

Q4. Describe the steps of digital image classification and how its accuracy is assessed.

Pre-processing, enhancement, supervised and unsupervised classification; confusion matrix, overall, producer's and user's accuracy, kappa; worked example 3.

Q5. Explain the working of GNSS and its applications in agriculture.

Distance from signal travel time, four satellites (Figure 3); GPS, GLONASS, Galileo, BeiDou, NavIC; DGNSS and RTK; uses.

Q6. Describe GIS data models and operations, and discuss applications of geospatial technology in Indian agriculture.

Figure 4 and table; coordinates and scale; Figure 5 and operations table; applications table — FASAL, drought, YES-TECH, AgriStack, precision farming.

8. Glossary

Term	Meaning
Attribute	Descriptive data linked to a spatial feature
Confusion matrix	Table comparing classified map classes with ground truth
Crop-cutting experiment	Harvesting and weighing a small random plot to estimate yield
GIS	Geographic information system
GNSS	Global navigation satellite system
Ground truth	Field observations used to train or check image interpretation
Kappa	Accuracy statistic that removes agreement expected by chance
Mixed pixel	A pixel containing more than one land-cover type
NavIC	Navigation with Indian Constellation — India's regional navigation system
NDVI	Normalised Difference Vegetation Index
NIR	Near-infrared radiation
Pixel	Smallest unit (cell) of a raster image
Raster	Data model of a grid of cells
RTK	Real-time kinematic correction giving centimetre-level GNSS accuracy
SAR	Synthetic aperture radar — an active microwave sensor
Spectral signature	Pattern of reflectance of a surface across wavelengths
Vector	Data model of points, lines and polygons
WGS 84	World Geodetic System 1984 — the datum used by GPS

References

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All diagrams and charts are original; the spectral curves in Figure 2 are schematic. Sensor specifications are from the cited agency pages. Values in the worked examples are illustrative, and every result shown was recalculated.

Practice MCQs

Q1. Healthy green vegetation reflects most strongly in the

(a) blue band
(b) red band
(c) near-infrared band
(d) thermal band

Q2. NDVI is calculated as

(a) (Red − NIR) ÷ (Red + NIR)
(b) (NIR − Red) ÷ (NIR + Red)
(c) NIR × Red
(d) (NIR + Red) ÷ 2

Q3. The size of the ground area represented by one pixel is the

(a) spectral resolution
(b) spatial resolution
(c) temporal resolution
(d) radiometric resolution

Q4. Which is an active remote-sensing sensor?

(a) Optical camera
(b) Thermal scanner
(c) Synthetic aperture radar
(d) Human eye

Q5. The finest spatial resolution of Sentinel-2 is

(a) 5.8 m
(b) 10 m
(c) 23.5 m
(d) 56 m

Q6. The spatial resolution of the LISS-IV sensor on Resourcesat is

(a) 5.8 m
(b) 10 m
(c) 23.5 m
(d) 56 m

Q7. NavIC is designed with a constellation of

(a) 4 satellites
(b) 7 satellites
(c) 24 satellites
(d) 31 satellites

Q8. A GNSS receiver needs signals from at least how many satellites for a 3D position?

(a) 2
(b) 3
(c) 4
(d) 6

Q9. A GIS data model that represents features as points, lines and polygons is

(a) raster
(b) vector
(c) grid
(d) pixel

Q10. Creating a zone of fixed width around a well in GIS is called

(a) overlay
(b) buffer
(c) interpolation
(d) classification

Q11. The accuracy of a classified crop map is assessed using a

(a) confusion matrix
(b) flowchart
(c) histogram only
(d) pie chart

Q12. YES-TECH uses satellite and weather data to estimate yields for

(a) crop insurance under PMFBY
(b) soil-health cards
(c) mandi prices
(d) land registration

Answers

Q1 — (c) near-infrared band
Q2 — (b) (NIR - Red) ÷ (NIR + Red)
Q3 — (b) spatial resolution
Q4 — (c) Synthetic aperture radar
Q5 — (b) 10 m
Q6 — (a) 5.8 m
Q7 — (b) 7 satellites
Q8 — (c) 4
Q9 — (b) vector
Q10 — (b) buffer
Q11 — (a) confusion matrix
Q12 — (a) crop insurance under PMFBY